

Ayesha Attiq

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Summary

Final-year chemistry undergraduate at NUST working at the intersection of electrochemistry and energy materials — electrospun nanofibers and artificial solid electrolyte interfaces for zinc metal batteries. Seeking research placements in energy storage and materials chemistry.

Education

National University of Sciences & Technology (NUST)

Bachelor of Science in Chemistry

KIPS College Islamabad

FSc Intermediate, Pre-Medical

Shining Star Public School

Matriculation, Science

Islamabad, Pakistan

Sept. 2023 – June 2027

Islamabad, Pakistan

Sept. 2019 – June 2021

Rawalpindi, Pakistan

Sept. 2015 – June 2019

Research Experience

AEMS Lab, USPCAS-E, NUST | Certification

Research Intern — Nanomaterials & Electrospinning

- Fabricated MoTe_2 nanostructured fibers by electrospinning; ran synthesis and materials characterization
- Analysed characterization data and reported results in written technical reports

June 2026 – Sept. 2026

Islamabad, Pakistan

Department of Chemistry (SNS), NUST | Certification

Research Intern — Battery Electrochemistry

- Synthesized polymer-based artificial SEI coatings to suppress zinc dendrite growth in zinc metal batteries
- Characterized cells by cyclic voltammetry, impedance spectroscopy (EIS), and galvanostatic charge-discharge

Aug. 2026

Islamabad, Pakistan

PINSTECH | Certification

Research Intern — Analytical & Materials Chemistry

- Ran electrochemical and spectroscopic analysis: cyclic voltammetry, UV-Vis, AAS, HPLC, GC, and GC-MS
- Developed and synthesized resin-based materials for electrochemical and catalytic applications

July 2025 – Aug. 2025

Islamabad, Pakistan

CETQAP | Certification

Intern — Quantum Computing for Chemistry

- Studied quantum computing and quantum algorithms through case studies and simulations
- Covered quantum state representation, entanglement, and data encoding for computational chemistry

July 2025 – Aug. 2025

Remote

Workshops & Training

MOSPI-2026 — Modeling & Simulation in Process Industries

DFT Cluster | SCME, NUST | Certification

- Introduced to density functional theory, electronic structure theory, and the VASP workflow
- Applied surface modelling to adsorption/desorption studies and catalytic reaction modelling

Sept. 2026

Islamabad, Pakistan

- Covered Li- and Na-ion batteries, supercapacitors, electrocatalysis, and energy storage materials
- Hands-on sessions on electrochemical characterization and electrochemical/biochemical sensors

Additional Coursework

Introduction to Process Safety and Risk Analysis | [Certification](#)

Introduction to HSE Engineering | [Certification](#)

Chemical Process Safety | [Certification](#)

Material Chemistry and Sustainable Building | [Certification](#)

Density Functional Theory | [Certification](#)

Petroleum Engineering with AI Applications — Specialization | [Certification](#)

Skills

Laboratory Techniques: Electrospinning & Nanofiber Fabrication, Nanoparticle Synthesis, Polymer Thin-Film Coating, Titrations, Chromatography, Gravimetric & Salt Analysis

Electrochemistry: Cyclic Voltammetry, Electrochemical Impedance Spectroscopy, Galvanostatic Charge-Discharge, Battery Performance Evaluation

Instrumentation: UV-Vis Spectrophotometry, Atomic Absorption Spectroscopy, HPLC, GC, GC-MS

Computational: Density Functional Theory (VASP), Origin, ChemDraw, Excel, High Scorer

Design: Photoshop, Illustrator

Social & Extracurricular

Deputy Director | *NLC, NUST*

- Managed media and event photography for NLC

Deputy Director | *SSC, NUST*

- Managed SSC events and media coordination

Executive Marketing | *NUST Robotics Club*

- Assisted marketing and outreach for NRC

Community Outreach & Volunteering

- Created a documentary focused on inflation and the well-being of the community as an outreach initiative
- Proactively involved in food drives, tutoring initiatives, and health surveys in underprivileged communities

Languages

English • Urdu